

Module 9

Implementing Security Management

Module Objectives

- Apply authentication, authorization, and accounting techniques to user management.
- Understand security controls, including managing network traffic.
- Assess vulnerabilities in cloud deployment scenarios.
- Monitor for suspicious activities.

Lesson 9.1

Identity and Access Management Concepts

Cloud+ CV0-004 Certification Exam Objective:

- 4.3 Given a scenario, implement identity and access management.

Identity and Access Management Concepts

- Identity and Access Management
- Secure Cloud Management Environment Access
- Secure Access to Cloud Resources
- Account Lifecycle Management
- Authorization Models
- Authentication Models
- Auditing Methods for Access Management

Identity and Access Management

- What is identity and access management?
 - Authentication
 - Authorization
 - Auditing
- IAM policies

Secure Cloud Management Environment Access

- Cloud web portals
- Common Language Infrastructure (CLI)
- Programmatic access to cloud resources
 - Application Programming Interface (API)
 - Software Developer Kits (SDK)

Secure Access to Cloud Resources

- Secure Shell (SSH)
 - What is key-based authentication?
 - Generate an SSH key pair
- Remote Desktop Protocol (RDP)
- Bastion host

Account Lifecycle Management

- Process
 - Creation
 - Management
 - Updates
- Deprovisioning
- Benefits
- Example (Azure)

Authorization Models

- Physical and logical access controls
- Mandatory access control (MAC)
- Discretionary access control (DAC)
- Group-based access controls
- Role-based access control
- Open Auth 2.0 Authorization (OAuth2)

Authentication Models

- Local user authentication
- Directory-based authentication
- Federated authentication
- Security Assertion Markup Language (SAML) authentication
- OpenID Connect
- Token-based authentication
- Multifactor authentication
- Single sign-on

Auditing Methods for Access Management

- Audit trail for authentication and authorization events
- Review accounts, groups, and roles

Lesson 9.2

Security Controls in the Cloud

Cloud+ CV0-004 Certification Exam Objective:

- 4.5 Given a scenario, apply security controls in the cloud.

Security Controls in the Cloud

- Endpoint Protection Controls
- Data Loss Prevention
- Intrusion Prevention System Intrusion Detection Systems
- Distributed Denial-of-Service Protection
- Firewall Controls
- Web Application Firewall Security
- Network Security Group

Endpoint Protection Controls

- Host-based intrusion detection and intrusion prevention systems
- Software firewall
- Log and event monitoring
- File integrity

Data Loss Prevention

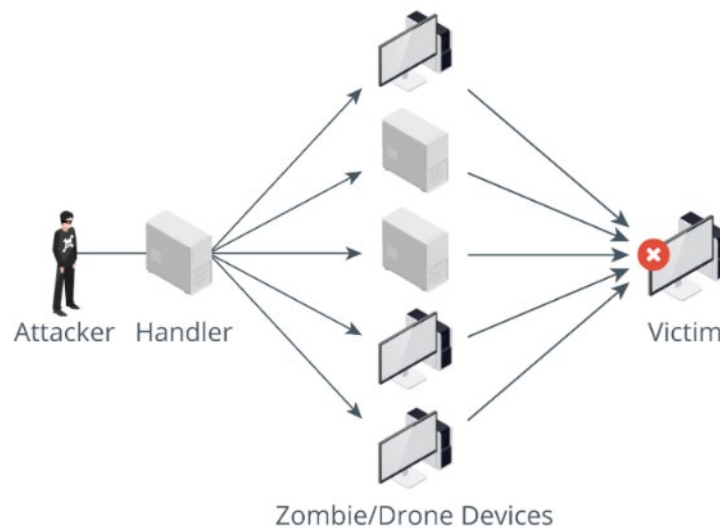
- Definition
- Key goals
 - Identify and understand the use of confidential data (in use, in storage, in transit).
 - Apply automatic protection to data using technology.
 - Monitor, detect, and respond to exfiltration attempts.
- Solutions and tools

Intrusion Detection Systems (IDS)

Intrusion Prevention System (IPS)

	Response	Traffic Flow
IDS	• Logs and alerts administrators • Threat detected: Logs and alerts	• Not inline with traffic flow • False positive: No impact on traffic
IPS	• Logs, alerts, and blocks threats • Threat detected: Logs, alerts, and blocks	• Inline and capable of filtering traffic • False positive: Blocks legitimate traffic

Distributed Denial-of-Service Protection



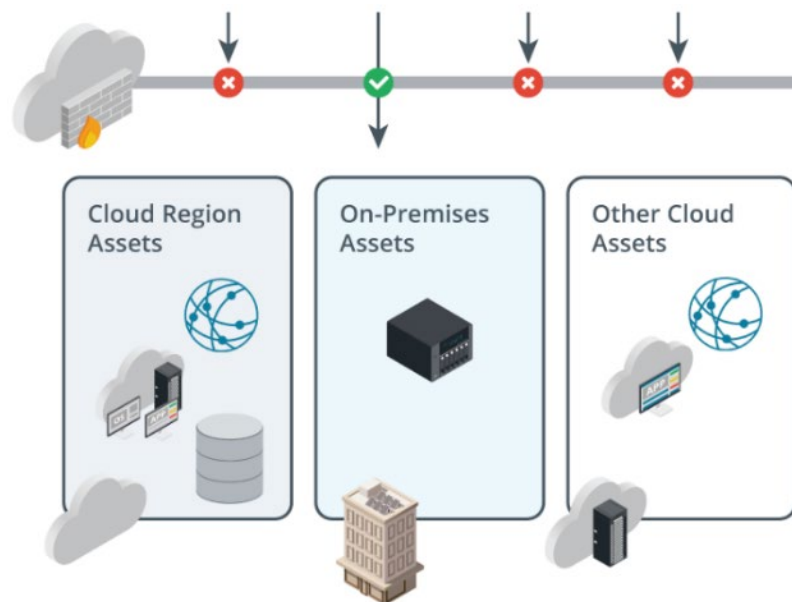
DDoS attack flooding the victim server. (Images © 123RF.com)

Firewall Controls

- Stateless firewalls
 - Network perimeter
 - Host device
 - Internal segments
- Network Access Control List (ACL)
 - Filter criteria
 - Logging

Web Application Firewall Security

- Two kinds of ACLs
 - Allow lists
 - Deny lists
- Virtual patching
 - Not controlling traffic
 - Configured virtual patch is not working



A web application firewall protects a company's cloud and on-premises assets. (Images © 123RF.com)

Network Security Group

- Definition
- Key features
- Comparison
 - Security groups
 - WAFs
- Holistic security approach

Lesson 9.3

Vulnerability Management Concepts

Cloud+ CV0-004 Certification Exam Objective:

- 4.1 Explain vulnerability management concepts.

Vulnerability Management Concepts

- Vulnerability Management
- Vulnerability Management Steps
- Common Vulnerabilities and Exposures

Vulnerability Management

- Default and common credentialed scans
- Agent-based scans
- Service availability

Vulnerability Management

1. Vulnerability scanning
2. Cloud service provider vulnerability scanners
3. Google Cloud platform container analysis
4. Amazon Web Services Amazon inspector
5. Azure Defender and Azure security center
6. Third-party vulnerability scanners:
 - Burp suite
 - Nessus
 - Nmap
7. Vulnerability identification
8. Vulnerability assessment
9. Vulnerability remediation

Common Vulnerabilities and Exposures

- Definition
- Components of a CVE entry
- National Vulnerability Database (NVD)

Score	Description
1.1+	Low
4.0+	Medium
7.0+	High
9.0+	Critical

Lesson 9.4

Monitoring Suspicious Activities to Identify Common Attacks

Cloud+ CV0-004 Certification Exam Objective:

- 4.6 Given a scenario, monitor suspicious activities to identify common attacks.

Monitoring Suspicious Activities to Identify Common Attacks

- Vulnerability Exploitation
- Social Engineering and Malware
- DDoS, Cryptojacking, and Zombies
- Metadata
- Responding to Security Incidents
- Event Monitoring
- Baseline Deviation
- Server Ports
- Unnecessary Open Ports

Vulnerability Exploitation

- Attack types vulnerability exploitation
- Human error
- Outdated software

Social Engineering and Malware

- Social engineering
- Phishing
- Malware
- Ransomware

DDoS, Cryptojacking, and Zombies

- Distributed Denial of Service (DDoS)
- Cryptojacking
- Zombie resources and instances

Metadata

- File metadata
- Metadata and social engineering

Responding to Security Incidents

- Data exfiltration or breach
- Website defacement
- Distributed Denial of Service (DDoS)
- Virus

Event Monitoring

- Purpose
- Examples of changes
- Tools
 - Azure Monitor
 - AWS CloudWatch
- Key features

Baseline Deviation

- Definition
- Example baseline standards
- Detection and remediation
- Tools
 - Automated configuration management
 - Baseline management

Server Ports

Protocol	Port	Description
HTTP	80	Unencrypted web connections
HTTPS	443	Encrypted web connections
RDP	3389	Remote desktop connections
SSH	22	Remote command-line connections
DNS	53	Name resolution services

Unnecessary Open Ports

- Default to all ports closed; open only necessary ports for specific server roles
- Configuration management scripts and utilities
- Tools example
 - Nmap

Module Summary

- Three A's: Authentication, authorization, and accounting manage user access.
- Security Controls: Examine connection sources, destinations, protocols, and configurations.
- Vulnerability Management: Conduct scanning and mitigation to protect against attacks.
- Best Practices: Establish vulnerability management, event monitoring, and maintain baseline configurations.